

CS with Fred

The Path to Architectural Mastery 2026

Phase 01 — Foundations

- C
- Memory Management
- Big O
- Data Structures

Phase 02 — Low-Level

- OS Kernels
- Assembly
- Concurrency
- Computer Organization

Phase 03 — Distributed

- Networking
- Microservices
- CAP Theorem
- Raft/Paxos

Phase 04 — Intelligence

- Neural Nets
- Vector Databases
- LLM Architecture

The Lab — Starter Projects

- Building a custom Memory Allocator
- Writing a Mini Operating System Kernel
- Implementing a Distributed Key-Value Store